Symposium Proposal:
Structuring Team Knowledge: Dimensions, Beliefs, Distribution, & Coordination

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Structuring Team Knowledge: Dimensions, Distribution, Coordination, & True Belief

Abstract

Understanding the structure of knowledge employed by organizational teams – how it is held, processed, or formatted for use – is crucial in fulfilling their promise for today's organizations. While many have proposed dimensions of knowledge, much of the work in this area has described the content of the knowledge (e.g., the domain and its measure), and therefore has not sufficiently accounted for the additional process concerns, such as coordination, directory systems, and distribution of knowledge. The presenters in this symposium offer several perspectives to inform and extend the structural conceptualization of team knowledge, including methodological considerations and connections to team performance outcomes. First, "Network Structure & Expertise Coordination in Knowledge Teams" discusses the coordination of specific expertise structures – networks of technical, design, and domain experts – and their effect on team performance. Next, "Knowledge, Ignorance, & Distortion: An Expanded Representation of Belief Structures in Groups" discusses the structure of belief within a team context, contrasting knowledge that is relevant in completing a given team task with other beliefs that teams may have (distortion) or lack (ignorance) in relation to the task. "Transactive Memory: Measuring Knowledge Distribution & Effectiveness" decomposes transactive memory into "sharedness," "accuracy," and "validation" dimensions. Finally, "Structural Dimensions of Team Knowledge" organizes these and other views of team knowledge into six key structural dimensions used in extant team knowledge research: sharedness, externalization, maturity, distribution, time-span, and tacit versus explicit. Together, these perspectives inform the domain of acceptable measures of team knowledge and advance understanding of its forms.

Key words: team knowledge, mental models, knowledge structure

Defining Team Knowledge: Dimensions, Structure, and Measures

As teams are increasingly utilized to process organizational knowledge, our ability to realize their full benefit is dependent on our understanding of how they hold and process knowledge (Anand et al., 2003; Cohen & Bailey, 1998). This goal necessitates both an understanding of the relevant dimensions of team knowledge and development of measures to assess their influence. In this symposium, we offer a set of perspectives that describe such dimensions, offer means of assessment, and connect to team performance outcomes.

At the individual level, knowledge has been defined as a broad concept "learned from experience or study… includ[ing] insights, interpretation, and information" (Schultz, 2001, p. 662) that varies in its readiness "to apply to decisions and actions" (Davenport, De Long, & Beers, 1998, p. 43). Many dimensions of such knowledge have been posited, including tacit vs. explicit (Polyani, 1956), externality, segmentation (Anand et al., 2003), maturity (Berman et al., 2002), dynamism/situational awareness (Wellens, 1993), accuracy (Cooke, Salas, Cannon-Bowers, & Stout, 2000), stickiness (Szulanski, 2000; von Hippel, 1994), and leakiness (Brown & Duguid, 2001). When considering knowledge at the team level, structural considerations add a layer of complexity to this dimensionality. First, knowledge is distributed across team members effectively forming a knowledge network with multiple knowledge sources that members can access. Second, some knowledge is individually held while other knowledge is shared thus creating knowledge structures that are more complex than the sum of the parts. How knowledge is distributed and shared can have a substantial effect on how the team interacts and operates. Therefore, team knowledge must account for process concerns, such as coordination, directory systems, and distribution of knowledge as more individuals are involved in the task environment.

Previous investigations have provided several descriptors of team knowledge structure

(Mohammed & Dumville, 2001). For instance, team mental models are defined as organized knowledge that team members share about the task and about other members that operate at or below the conscious level, acquired by working, interacting and training together, which helps members form accurate explanations and expectations about task activities (Cannon-Bowers, Salas, & Converse, 1993; Klimoski et al., 1994; Kraiger & Wenzel, 1997). Mutual knowledge, on the other hand, is understood to be knowledge that team members share and know that they share it, improving common ground among team members and making their communication more effective (Clark & Marshal, 1981; Cramton, 2001; Krauss & Fussell, 1990). Understanding among team members of the processes and technologies that affect their mutual performance is termed shared knowledge (Nelson & Cooprider, 1996). Collective mind has been described as individual team members having an understanding of the group's task and of each other, and how this contributes to the work of the group (Weick et al. 1993). Finally, transactive memory has been described as knowledge of who knows what in the team, which helps teams determine who to contact in order to access others' individual knowledge and to assign task responsibilities to the respective specialists in the team (Wegner, 1986).

Collectively, these various perspectives of team knowledge offer rich explanations of how knowledge operates in teams. At the various definitions make it difficult to formulate a consistent representation of team knowledge, which raises a number of questions and presents a number of challenges to develop measures and compare results across studies. One important question is, do we need a unified and consistent representation of team knowledge? Or is it better to have these multiple representations and use the one best suited to a particular research problem? We argue that it is possible to reconcile these multiple perspectives into a multi-dimensional representation of team knowledge that describe its various structural aspects.

The first three individual presentations in this symposium explore various structural aspects, extending our understanding of how knowledge is held within a team. First, Faraj ("Network Structure & Expertise Coordination in Knowledge Teams") presents a perspective of team knowledge as expertise networks and discusses the coordination of specific expertise structures – networks of technical, design, and domain experts – and their effect on team performance. Next, Anand, Gupta, and Henderson ("Knowledge, Ignorance, & Distortion: An Expanded Representation of Belief Structures in Groups") discuss the structure of belief within a team context, contrasting knowledge that is truly of help in completing a given team task with other beliefs that teams may have (distortion) or lack (ignorance) in relation to the that task. Then, Gupta and Hollingshead ("Transactive Memory: Measuring Knowledge Distribution & Effectiveness") decompose transactive memory into "sharedness," "accuracy," and "validation" dimensions. Finally, Clark and Espinosa ("Structural Dimensions of Team Knowledge") organize these and other various views of team knowledge into six key structural dimensions that have been used in extant team knowledge research: sharedness, externalization, maturity, distribution, time-span, and tacit versus explicit.

In addition to discussing its dimensions, our presenters seek to improve the methods used to measure team knowledge structure. To date, there has been little consensus on a method for assessing and measuring knowledge used by teams. Some have assessed content domains either directly (Austin, 2003; Cooke, 2003; Espinosa, Carley, Kraut, Lerch, & Fussell, 2002; Mathieu, Goodwin, Heffner, Salas, & Cannon-Bowers, 2000), or through proxies such as textual analysis of participant summaries (Boland et al., 2001; Carley, 1997) and conversations (Kuhn & Corman, 2003), through shared experience of NBA teams representing tacit knowledge (Berman et al., 2002), shared experience of project teams (Stasser, Stewart, & Wittenbaum, 1995), and educational and functional backgrounds of team members (Van der Vegt et al., 2003). Others have measured ratings of knowledge held – observations, self-reports and agreement of members about knowledge held. For instance, Devine (1999) had observers rate instances of information similarity, while Ensley & Pearce (2001) measured self-reports about individuals and team knowledge and Lewis (2003) devised a measure rating whether measures believe their teammates to hold relevant knowledge. Each of these approaches measure some form of knowledge similarity (comparing stocks of knowledge) and attempt to demonstrate what part of knowledge is shared. However, none account for the distribution of knowledge within a team or the role of individual unshared knowledge as a component of total team knowledge.

Our presenters address these important issues. Faraj proposes measuring the distribution of different types of expertise, including the role of the team leader in the internal knowledge network. Gupta and Hollingshead specify types of distribution as components of a transactive memory system – sharedness, accuracy, and validation – to guide measures of knowledge location and differentiation. Anand and his colleagues stress the importance of understanding how such a distribution of knowledge rests on the "true knowledge" needed to complete a task. Clark and Espinosa include both sharedness and distribution as primary structural dimensions essential to understanding team knowledge as a construct. Together, these perspectives inform the domain of acceptable measures of team knowledge and advance understanding of its forms. Our symposium discussant, Prof. Salas, a senior scholar with substantial research expertise in team cognition, will discuss the similarities and differences among these views and in an attempt to provide a perspective that reconciles these multiple perspectives of team knowledge into a coherent perspective to guide future research on team knowledge.

Relevance of the Symposium to AOM Divisions

This symposium offers a forum to discuss structural aspects of team knowledge and their measurement issues. The topic is of interest to a variety of Academy divisions. Knowledge research is at the intersection of many disciplines, from information technology and operations management to organizational theory and behavior. The meso perspective inherent in the team context speaks to the core of the Organizational Behavior division, particularly when adding the connection to performance. Those in the Organizational Communication & Information Systems division will be interested in the implications of knowledge measurement for the design of IT software and communication systems. Finally, the attempt to develop the team knowledge construct through its dimensionality and measures should appeal to the Managerial & Organizational Cognition division.

Network structure and expertise coordination in knowledge teams
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Research on the coordination of organizational teams has long favored an organizational design perspective where coordination is achievable through the proper matching of coordination structures to environmental demands. Previous research on coordination (e.g., Gupta, Dirsmith, & Fogarty, 1994; Van de Ven, Delbecq, & Koenig, 1976) has emphasized the mode of coordination (e.g. programmatic versus improvisational mechanisms) or the frequency of communication (a proxy for information processing capacity). However, it may be just as important to focus on the content of coordination (what is being coordinated). Further, more and more organizations are operating on the high end of the traditional coordination needs scales and are extensively using all available modes of coordination. Yet, even when facing similar work environments and having access to similar coordination mechanisms, some teams and work units achieve superior coordination compared to similar others (Gittell, 2001). One reason may be that complex knowledge work requires the application of specialized skills and knowledge in a timely manner and thus raises difficult coordination issues in dynamic environments (Edmondson, 1999; Faraj & Sproull, 2000; Gittell, 2002). In the final analysis, we still know little about how teams coordinate their knowledge work, especially under demanding task environments.

To better understand the work of knowledge teams, researchers have recently proposed to view the team from an explicit information exchange and integration perspective (Hinsz, Tindale, & Vollrath, 1997). This emergent perspective opens new possibilities regarding the investigation of how team expertise is coordinated (Faraj & Sproull, 2000; Gittell, 2002), how tacit task aspects affect outcomes (Subramanian & Venkatraman, 2001), and the impact of shared mental models (Mohammed & Dumville, 2001) or mutual knowledge (Cramton, 2001). The network perspective has recently provided a complementary perspective for the study of knowledge work teams. Using the tools of social network analysis, a few studies have found that teams where interactions are hierarchical or where the leader is positioned as a structural hole within the team's communication network exhibit reduced ability to coordinate and thus perform less well than other teams (Cummings & Cross, 2003; Reagans & Zuckerman, 2001).

How existing expertise networks affect expertise coordination is the focus of this presentation. Specifically, I will report findings based on a cross-sectional study of 69 software development team in one large organization. Teams were included in the study if they were developing business application software in order to keep the task comparable. Teams had to be of medium size (about 10 full-time members) in order to control for the different dynamics and processes that characterize very small or large teams. Only teams that had completed their requirements definition stage were selected in order to eliminate teams that were still in formation or struggling to define their task. Finally, I did not include teams that had completed their projects in order to avoid retrospection bias. These stringent selection criteria allow an unusual control over task uncertainty and team size both factors that can impact team performance.

For this symposium, I will focus on the expertise coordination processes that are operating within the team as well as the network structure that exists based on team interactions. Specifically, I will provide comparative assessments of 3 expertise networks: 1) the technical expertise network (the people who know a great deal about specialized technical areas); 2) the design expertise network (the people who know a great deal about software design principles and architecture); and 3) the domain expertise network (the people who know a great deal about application domain and client operations). The measures of expertise coordination processes and team performance were collected through questionnaires to the team members and the non-team stakeholders respectively.

A first set of questions to be addressed in this presentation relate to the link between network structure and team outcome and performance variables. Does having the team leader occupy a central position in the network lead to better outcomes? This is what research in software development has shown historically (Baker, 1972; Schneiderman, 1980). Or, will the opposite be true as shown recently in knowledge teams (Cummings & Cross, 2003). A second set of questions relates to the distribution of expertise on the team. Will teams with a well developed core-periphery structure be more effective than ones that are characterized by diverse ties? Finally, how do the expertise network patterns affect team-level expertise coordination, a factor that has been shown to be crucial for performance (Faraj & Sproull, 2000)?

A few recent studies have empirically linked structural properties of within team network to its performance. Such studies have focused primarily on the communication structure (Cummings & Cross, 2003; Reagans & Zuckerman, 2001) or on advice networks (Sparrowe, Liden, Wayne, & Kraimer, 2001). The study I am reporting on extends existing knowledge in significant ways: first, it is focused on the distribution of expertise within the team; second, it differentiates between three types of expertise (technical, design, and domain) and thus allows the evaluation of how different expertise networks overlap and if their impact is mutually reinforcing; and third, the measurement of both the expertise network as well as the expertise coordination processes allows the comparison of the impact of team structure (i.e., the expertise network) versus the relevant team processes (expertise coordination) on team performance.

Overall, I am suggesting that the investigation of expertise networks within a team provides a significant addition to team research that has privileged the knowledge or expertise dimension of team work. As we move away from the traditional socio-psychological perspective on teamwork to one that emphasizes team knowledge, much is gained by investigating network effects.

Knowledge, Ignorance & Distortion: An Expanded Representation of Belief Structures in Groups
Vikas Anand, Nina Gupta, Kevin Henderson,
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Knowledge, Ignorance & Distortion: An Expanded Representation of Belief Structures in Groups

Organizations increasingly use work groups as vehicles for performing knowledge work. In recent years, a significant body of research has focused on understanding how knowledge is stored and used in groups (Anand, Clark, & Zellmer-Bruhn, 2003; Cicourel, 1990; Darr, Argote, & Epple, 1995; Hoopes & Postrel, 1999; Wegner, Erber, & Raymond, 1991; Wegner, Guiliano, & Hertel, 1985). Most researchers, depict a group's knowledge in terms of the beliefs (and corresponding belief structures) collectively held by its members. In this model, beliefs that are 'true justified' (either by empirical justification or through reasoned logic) constitute knowledge (Nonaka, 1994; Nonaka & Takeuchi, 1995; Polanyi, 1958; Shope, 1983).

A belief-based representation has been useful in advancing our understanding of how groups store and use knowledge. However, researchers often adopt incomplete representations because they ignore the presence of unjustified beliefs in group knowledge structures. Additionally, researchers have not focused on the 'ignorance' of the group – where ignorance is defined as the absence of beliefs needed for completion of the group's tasks. In this paper, we introduce a more complete representation of a group's knowledge – one that simultaneously considers all three types of beliefs: justified, unjustified, and the missing.

Unjustified beliefs, while technically not constituting knowledge, play a big role in determining group outcomes. For instance, a compensation committee may erroneously believe that a certain type of pay incentive is an ideal employee-motivator. Even though this belief is not really a part of the group's knowledge (because it is an unjustified belief), it plays a key role in determining the group's decision. Similarly, ignorance of key beliefs that are ideally needed to perform a group's task can also be an important determinant of group outcomes.

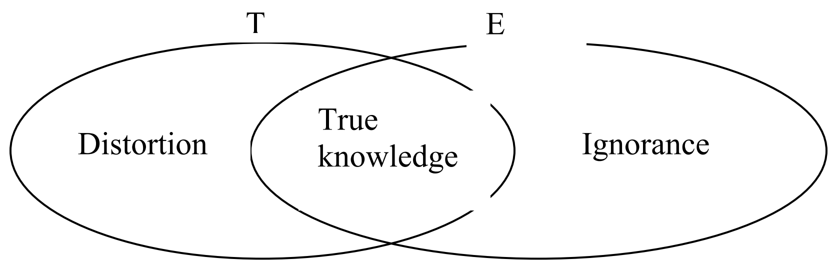
Unjustified and absent beliefs are especially significant because, in addition to their direct effects, they likely affect group outcomes through their interactions with each other and with justified beliefs. For instance, consider two groups, A and B, that have similar levels of justified beliefs. Group B, however, has a large amount of unjustified beliefs, while group A does not. Despite their similar levels of justified beliefs, their performance (in terms of effectiveness of their decisions) is likely to vary in the near and long term. Since group A is unfettered with erroneous beliefs it is likely to outperform B in the short term. However, the poor consequences initially faced by B are likely to spur a reexamination of their currently held beliefs. This process may result in learning that can change the group's belief structures (decrease in unjustified beliefs and increase in justified beliefs) that may improve future performance. Alternately, since powerful group's often enact and shape their own environments, group B may influence and modify its environment to be consistent with its originally distorted beliefs (Weick, 1995).

Group Knowledge Structures: Knowledge, Distortion and Ignorance

Figure 1 depicts an expanded model of a group's knowledge structure that addresses the above issues. In the figure, set T represents all beliefs that group members hold about the focal task, and the set E is an idealized set that contains **all** possible **true** beliefs that are required to perform a task. The intersection of the two sets represents the group's **true knowledge** – those of its beliefs that are relevant to the task at hand and are justified. The intersection of T' and E ($T' \cap E$) represents a set of justified true beliefs that are needed to perform the task properly, but are not possesses by the entity. Thus $T' \cap E$ represents the **ignorance** of the group. On the other hand, the intersection of E' and T ($E' \cap T$), represents the group's set of unjustified beliefs.

These beliefs represent a world-view that is inconsistent with reality and we term them **distortion**.

Figure 1: Distortion, Knowledge and Ignorance



Note that sets T and E are dynamic. Set T is constantly changing as a function of group learning and group turnover, while set E is constantly changing as a function of continuous change in the task environment. Incorporating the dynamic nature of these sets in a model of group knowledge provides us a richer and more realistic understanding about group knowledge processes.

Implications for group learning and performance

Our paper describes the above model of group knowledge structures and teases out its implications for two group outcomes – performance and group learning. We look at how different configurations of a group's knowledge affect diverse performance outcomes such as speed of decision making, decision quality etc at different points in time. We focus on group learning for two reasons: First, in today's knowledge based environments, continuous learning is a critical predictor for long term success – a group needs to learn to maintain its knowledge in the face of a changing task environment (Fiol & Lyles, 1985; Garvin, 1993; Huber, 1991), and second, the group's learning process alters its knowledge structure – thus as a result of learning its distortion and ignorance may be replaced by knowledge (or vice versa).

A variety of propositions suggest themselves and we list a couple of them. For instance, a group that operates primarily on the basis of true knowledge, with little ignorance and distortion, has relatively low motivation to try new innovations that spur further learning (Christensen, 1997; Christensen & Bower, 1996). Indeed, as (Miller, 1993)) points out, even groups which have achieved success through continuous learning, reduce their learning as they move into a 'maintenance' mode. This suggests that such a configuration could lead to superior short term performance that declines over time.

On the other hand, a group with large amounts of distortion and knowledge receives mixed feedback from the environment in the form of ambiguous results. Such a group is likely to perceive a tremendous amount of equivocality and uncertainty in its environment (Wiseman & Gomez-Mejia, 1998). A key approach taken by groups to reduce uncertainty is through additional information seeking (McCrimmon & Wehrung, 1986). The impact of this new information obtained on the group's knowledge structure depends on the amount of within-group discussion that accompanies its acquisition. A group's knowledge likely increases if it discusses and evaluates newly acquired information. On the other hand if incoming information is accepted without discussion and deliberation, the group's distortion and knowledge may both increase (Anand, Glick, & Manz, 2002; Anand, Manz, & Glick, 1998; Daft & Weick, 1984).

Conclusion

We propose a new structure which represents group knowledge in terms of all relevant beliefs – justified, unjustified, and those that are missing. We will develop this further by exploring different knowledge configurations that groups possess and examine how they impact the group's performance and learning rates over time.

Transactive Memory Perspective on Team Knowledge Measures: A Review & Agenda

Naina Gupta, Andrea B. Hollingshead
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Organizations today face an environment that is rapidly changing both due to quick changes in technology as well as constant availability of vast amount of information. Frequently, managers use workgroups to increase an organization's knowledge processing capabilities. Groups working over time, often, develop a systematic way of processing knowledge known as a transactive memory system.

Transactive memory refers to the idea that team members often develop a shared division of cognitive labor with respect to the encoding, storing and retrieving of knowledge from different substantive domains (Wegner, Giuliano, & Hertel,1985; Hollingshead, 2001). Team members divide the labor (responsibility of knowledge processing in different domains) based on their shared awareness of the knowledge distribution (or "who knows what") in the group. Research suggests that transactive memory systems develop naturally when members are interdependent (Hollingshead,2001) as they often are in work teams (Liang, Moreland, & Argote, 1995). Each team member becomes a specialist in some domains but not others; and members rely on one another to access information from appropriate domains (Hollingshead,2001). Specialization reduces the cognitive load on each individual but allows individuals access to more information collectively. The shared awareness of the knowledge distribution enables the team members to better coordinate their actions (Cannon-Bowers, Salas, & Converse,1993).

Transactive memory considers both the shared awareness of the knowledge distribution amongst the team members and the specialized knowledge held by each team member as team level knowledge. This consideration of shared knowledge from a complementary perspective (knowledge divided among team members) over and above the overlapping perspective adopted in much of team knowledge research, provides a more holistic understanding of the construct (Mohammed & Dumville, 2001).

In this presentation, we will provide a framework and review of the existing measures of transactive memory systems. We will conclude the presentation with data from a study that seeks to address gaps in the current transactive memory system measures.

Dimensions of Effectiveness

To organize our discussion of the dimensions of an effective transactive memory, we consider a framework introduced by Brandon & Hollingshead (2004). The three dimensions are sharedness of the perceptions of the transactive memory among the team members, the accuracy of these perceptions and the validation of these perceptions (Brandon & Hollingshead,2004). Sharedness refers to the extent to which all the members of the team have similar perceptions on 'who knows what' in the team (Lewis,2003; Brandon & Hollingshead,2004). Accuracy refers to the extent to which these perceptions reflect the actual knowledge of those team members (Austin,2003; Brandon & Hollingshead,2004). Validation of the transactive memory refers to the extent to which team members participate in the area of the group task for which they are held responsible by the team members (Brandon & Hollingshead,2004). In other words, the degree to which there is a match of 'who does what' with 'who knows what' in the team. Transactive memory will be most effective when there is complete sharedness, accuracy and validation of team member perceptions of expertise in the group.

These dimensions have been examined in both the laboratory and the field. The unit of analysis in laboratory research has ranged from couples (Wegner, Raymond, & Erber,1991; Hollingshead,1998a, b, 2001) and coworkers (Hollingshead,2000) to work groups (Liang, Moreland, & Argote,1995; Moreland, Argote, & Krishnan,1998; Gupta,2005). Tasks in laboratory studies have included both cognitive tasks such as recalling words and pooling knowledge (Wegner et al.,1991; Hollingshead, 1998a, b, 2001) and production tasks such as assembling small electronic equipment such as radios and stereos (Moreland, Argote, & Krishnan,1996; Moreland & Myakovsky,2000; Lewis,2003)). The unit of analysis in field research has been workgroups such as MBA consulting teams, project, cross functional, and functional teams in technology companies (Lewis, 2003) and merchandizing teams (Austin,2003). Ongoing teams performed tasks varying from analyzing market, competitor and technological systems, determining purchasing and manufacturing levels, making new product recommendations to providing ongoing support (Lewis, 2003; Austin, 2003).

The extent of sharedness of transactive memory in the workgroup has been assessed by asking group members for their perceptions of relative expertise of members. Laboratory measures of sharedness include agreement between couples on the relative expertise in the different domains of knowledge (Wegner et. al, 1991, Hollingshead, 1998a, b, 2001), agreement across group members in their identification of experts in radio building skills (Moreland,1999). When team members agree on the relative expertise of all team members, there is a high level of sharedness in the transactive memory. The field measures for sharedness includes an agreement index called Quadratic Assignment Procedure (QAP) index and a consensus score. Both measures initially involve a similar process as described for the laboratory measures. However, they differ in that the QAP index uses a networks analysis program to combine the member-expertise matrices of all the group members (Lewis, 2003) and the consensus score involves the standard deviation of member expertise rating for each knowledge area (Austin,2003).

Laboratory measures of the extent of accuracy include linking perceptions of each member's expertise in a domain of knowledge with an objective measure of the members' performance in that domain in a knowledge pooling task (Hollingshead, 1998) or a radio assembly task (Moreland,1999). The field measure for accuracy is similar to that used in laboratory research except that it includes linking the team member identified experts with the experts' performance in problem solving scenarios (Austin,2003).

Few, if any, studies have directly measured the extent of validation among members in transactive memory systems. Much of the past research in transactive memory has examined sharedness and accuracy of the perceptions of expertise among team members to predict transactive memory effectiveness. The underlying assumption is that the experts will participate in the area for which they are responsible by either performing the subtask or providing the information (Moreland et al.,1998; Hollingshead,1998b). Direct measures of validation are required to test this assumption. One proposed measure of validation is an assessment of instances of display of expertise or sharing of knowledge in the area of expertise (Brandon & Hollingshead, 2004).

Moreover, few studies have examined transactive memory over a period of time (Lewis, Gillis, & Lange,2003). There are few, if any, studies that assess how groups respond to changes in their transactive memory systems, for example, in the case of membership change or when group members who do not share their unique knowledge with others in their group. A comprehensive measure encompassing all three dimension of transactive memory administered at intervals of time would enable an assessment of transactive memory changes. Future research should focus on the study of validation of transactive memory and of changes in transactive memory over time.

We will end the presentation by presenting data from a study that examines validation processes in transactive memory over time. (Data collection is currently underway; we expect to have findings to report at the 2005 AOM meeting.) Specifically, we are investigating how members make attributions and take action when other members do not make their expected knowledge contribution to their group. We vary the causes attributed to the lack of participation of a team member and examine the responses made by other group members. We predict that when group members attribute the lack of participation to ability rather than to effort, they will be less likely to judge the person harshly and more likely to help the low participator accomplish their task by providing knowledge or non-knowledge support. In both cases, we expect members to make adjustments to their transactive memory systems over time. We will discuss the implications of these responses for transactive memory change and effectiveness.

Structural Dimensions of Team Knowledge
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Efforts to understand team knowledge have led to multiple, sometimes conflicting, representations of the concept (Hammer, Leonard, & Davenport, 2004). Some have described dimensions of knowledge that relate to its content, as when Schultz (2001) defined knowledge as a broad concept "learned from experience or study… includ[ing] insights, interpretation, and information" (p. 662) in a given domain. Others have pointed out that knowledge varies in its structure, its readiness "to apply to decisions and actions" (Davenport, De Long, & Beers, 1998, p. 43); for example, whether individuals hold knowledge internally or simply have access to external sources (Anand, Manz, & Glick, 1998). In addition, individual knowledge is partially unshared, representing sources of unique expertise in a team, and partially shared, representing areas of knowledge overlap that helps teams coordinate and work cohesively as a team. Different combinations of individual and shared knowledge create knowledge structures that are more complex than the sum of the parts. This structural complexity is exacerbated at the team level, where considerations grow beyond individuals, and must encompass process elements. To help clarify the influence of this complexity, we offer a set of dimensions related to team knowledge structure (how the knowledge is held, processed, or formatted to be usable) – *sharedness, externalization*, *distribution*, *maturity, tacit vs. explicit*, and *time span*.

DIMENSIONS OF TEAM KNOWLEDGE STRUCTURE

Dimension 1: Shared vs. Individual. Team knowledge has typically been characterized as knowledge shared between members; however, we believe it is may be more accurately characterized as both the individual knowledge possessed by each team member that is not possessed by other members, as well as the knowledge that is shared among members or factions. Individual unshared knowledge can be most effectively used when other team members have some idea of the nature of the unshared knowledge, although teams whose members share fully in each others' knowledge bases may not be ideally suited for synergistic performance. Shared knowledge provides a common knowledge base for team interaction and the coordination of individual performance (Wittenbaum & Stasser, 1996).

The shared portion of team knowledge has been characterized in a number of ways, including: shared mental models – i.e., organized knowledge that team members share about the task and each other, acquired by working, interacting and training together, which helps members form accurate explanations and expectations about task activities (Cannon-Bowers, Salas, & Converse, 1993; Klimoski et al., 1994; Kraiger & Wenzel, 1997); mutual knowledge – i.e., knowledge that team members share and know that they share it, which improves the common ground among team members, making their communication more effective (Clark & Carlson, 1982; Clark & Marshal, 1981; Cramton, 2001; Fussell & Krauss, 1992; Krauss & Fussell, 1990); shared knowledge – that is, an understanding among team members for the technologies and processes that affect their mutual performance (Nelson & Cooprider, 1996), which provides members with the ability to access knowledge sources in the team and apply them effectively when necessary (Alavi et al., 2001); and transactive memory – knowledge of who knows what in the team – which helps teams determine who to contact when they to access others' individual knowledge when needed and assign task responsibilities to the respective specialists in the team (Wegner, 1986, 1995).

Dimension 2: Internal vs. External. In order to accomplish their task directives, teams may choose to make use of external knowledge sources in addition to the knowledge held by the members (internal to the team). Members depending on external resources do so through the use of some form of directory structure – a mental telephone book of sorts, listing knowledge resources outside the team, rather than by acquiring the knowledge for themselves (Anand, Manz, & Glick, 1998). Utilizing externalized knowledge means that members may never personally possess the knowledge, instead using a variety of techniques to employ external sources. For instance, a team may delegate a task to a non-member, or it may invite outsiders to specific team meetings to work on designated aspects of the team's task. Teams differ in the amount of external knowledge that they use, as each develops different norms about knowledge sources.

Dimension 3: Even vs. Distributed. The knowledge held by a team can also vary in its distribution among members or factions. be relatively evenly or relatively considered in terms of its distribution among the members: A team with relatively even distribution could have members who each hold expertise in separate fields relevant to the task, while uneven dispersion may be when relatively few members have distinct knowledge stores while the other members or factions hold similar stores of task-related knowledge. Uneven distribution, especially between factions, decreases the likelihood of knowledge-sharing across functional lines, as an in-group – out-group dynamic is introduced (Tajfel & Turner, 1985). As Faraj discussed above, knowledge distribution in a team can be represented as networks of various types of expertise present in a team.

Dimension 4: Emergent vs. Mature. Team knowledge content will also vary in its level of maturity (cf. Berman et al., 2002), characterized by its level of readiness for application to the performance objective (dynamic properties comprise a separate dimension, discussed below). Knowledge emerges as a product of the interaction between team members, although in such shared contexts emergent knowledge is not necessarily newly created. For instance, knowledge that is held only by one or more members may be transferred to others through exchange of ideas, information, or procedures, thus emerging in the group context. Other knowledge may represent novel combinations of previous information or new insights, thus emerging as created knowledge. In either case, this emergent knowledge may then be subject to influence from a variety of task and team factors and processes as it matures into knowledge that the group can agree upon. More mature knowledge solidifies into a cognitive form that has a certain amount of stability or institutional inertia, becoming a norm, plan, or a way of completing tasks.

Emerging knowledge may arise as a perspective or line of inquiry held by one or more team members, or as a combination of differing perspectives in the case of created knowledge. As it emerges, knowledge may be shaped by group process factors such as power dynamics and interpersonal diversity. The locus of power in a team may influence which aspects of knowledge will be accepted by the group, while diversity of backgrounds, values, and personality may thwart communication efforts (Zenger & Lawrence, 1989) and possibly create factions (Lau & Murnighan, 1998) which may impede knowledge transmission or influence its interpretation (Clark, Anand, & Roberson, 2000). More task-related aspects of the knowledge such as accuracy and functionality may also direct the development of emerging knowledge.

More mature knowledge will be less susceptible to influence from such dynamics and will generally have increased resistance to change or revision, sometimes to the point of "ossification" (Berman et al., 2002). Team members will be more familiar with both the explicit and tacit aspects of the knowledge, accepting or at least abiding by its character. Mature team-level knowledge is likely to be present in a team that has had significant time or experience together. At those later points in the lifecycle, members' stores of knowledge are likely to have been exposed to other team members, reducing the overall differentiation of the team (Gersick &

Hackman, 1990). At this point, team members may also have less need to communicate when utilizing knowledge or engaging in routine procedures.

Dimension 5: Tacit vs. Explicit. The content of team knowledge may be fairly explicit, detailed and communicated without much difficulty (e.g., an organization's recruitment manual), or more tacit or procedural, such as skill in servicing high tech equipment that is necessarily more time-consuming and difficult to transfer (Kogut & Zander, 1997). A simple division between tacit and explicit, however, belies what is often a strong association between the two knowledge forms. A recruitment manual will lay out organizational standards, for example, but there will most often be a great deal of less explicit, but related, knowledge that is necessary to actually complete the job. Learning the tacit aspects of a task most often requires some increased level of interaction between organizational members, ranging from simple observation to coaching and actively working together on tasks. Due to its inscrutable nature, tacit knowledge is more susceptible to being lost or incorrectly translated from member to member. It is especially vulnerable to the effects of turnover at the team or organizational level, or poor training and mentoring programs. When properly supported within a team or organizational environment, tacit knowledge may be less susceptible to decay than simple written or spoken instructions (Cohen, 1996). It also may be more difficult to track transmission of tacit knowledge. Organizations which have close relationships with suppliers and clients may have to create multi-organizational task forces to promote knowledge transfer, a resource-intensive solution which also risks knowledge hemorrhage to outside parties.

Dimension 6: Time Span. This dimension refers to the time span over which the knowledge is relevant to the task. Much of the team cognition literature deals with either long-lasting or fleeting knowledge, but rarely are both considered, despite their distinct natures

(Cooke et al., 2003). Long-lasting knowledge is acquired over time through experience with a task domain, and by interacting and training with other team members, and it is generally relevant for the task over long periods of time (e.g., knowledge of a software language). Representations of this include team mental models (e.g., Carley, 1997; Kraiger et al., 1997; Mathieu et al., 2000), schema similarity (Rentsch & Hall, 1994), knowledge similarity (Espinosa et al., 2002a); shared knowledge (Nelson et al., 1996), and transactive memory (Wegner, 1986). Team members also need a more immediate fleeting knowledge, which is relevant for specific task situations, thus it is short-lived. Most discussions of dynamic knowledge in the literature derive from "situation awareness", a concept that has been studied intensively in military and aviation contexts, defined as "up-to-the minute" knowledge necessary to interact with a human or physical system (Adams, Tenney, & Pew, 1995). Situation awareness in a team context can be defined as understanding how events in the task environment, such as the activities of other team members, affects the future of the task and one's own activities (cf. Dourish & Bly, 1992; Endsley, 1995). Thus, team members should not only have situational awareness to carry out their task responsibilities, but also team situation awareness to work effectively as a coordinated unit (Wellens, 1993).

One important property of awareness knowledge is that its relevance diminishes when the situation ends. For example, a pilot flying an airplane needs to be attentive to instrument readings and weather conditions during the flight, but once on the ground such information is no longer relevant and can therefore be forgotten. It also should be noted that the concept of situation awareness has been extrapolated to many aspects of team collaboration, including task awareness (Espinosa et al., 2000), workspace awareness (Gutwin & Greenberg, 1999; Gutwin & Greenberg, 2004); presence awareness (Boyer, Handel, & Herbsleb, 1998), and others

(Steinfield, Jang, & Pfaff, 1999). Our terms, long-lasting knowledge and dynamic knowledge, are similar to the categorization proposed by Cooke and her colleagues (2000), but more inclusive of the various types of knowledge content which may be represented.

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